This note derives the matrix $\boldsymbol{A}_{\mathrm{CB}}$ of `pose_estimation.pdf`, which carries the camera pose covariance over to the robot base.

Lemma 1. *Let the camera pose in the world frame be given by the rotation* ${}^{\mathrm{W}}\boldsymbol{R}_{\mathrm{C}}$ *and the position* ${}^{\mathrm{W}}\boldsymbol{p}_{\mathrm{C}}$*, and let the robot base be rigidly attached to the camera, with its origin at* ${}^{\mathrm{C}}\boldsymbol{o}_{\mathrm{B}}$ *and its orientation* ${}^{\mathrm{C}}\boldsymbol{R}_{\mathrm{B}}$ *in the camera frame, both known exactly. Let* $\delta\boldsymbol{p}_{\mathrm{C}}$ *and* $\delta\boldsymbol{\theta}_{\mathrm{C}}$ *be small perturbations of the camera position and orientation, expressed in the world frame, such that*

$$ {}^{\mathrm{W}}\tilde{\boldsymbol{p}}_{\mathrm{C}} = {}^{\mathrm{W}}\boldsymbol{p}_{\mathrm{C}} + \delta\boldsymbol{p}_{\mathrm{C}}, \qquad {}^{\mathrm{W}}\tilde{\boldsymbol{R}}_{\mathrm{C}} = (\boldsymbol{I} + [\delta\boldsymbol{\theta}_{\mathrm{C}}]_{\times})\,{}^{\mathrm{W}}\boldsymbol{R}_{\mathrm{C}}, $$

with covariance ${}^{\mathrm{W}}\boldsymbol{\Sigma}_{\mathrm{C}}$*. Then, to first order, the resulting perturbations* $\delta\boldsymbol{p}_{\mathrm{B}}$ *and* $\delta\boldsymbol{\theta}_{\mathrm{B}}$ *of the base pose, in the same convention, are*

$$ \begin{pmatrix} \delta\boldsymbol{p}_{\mathrm{B}} \\ \delta\boldsymbol{\theta}_{\mathrm{B}} \end{pmatrix} = \boldsymbol{A}_{\mathrm{CB}} \begin{pmatrix} \delta\boldsymbol{p}_{\mathrm{C}} \\ \delta\boldsymbol{\theta}_{\mathrm{C}} \end{pmatrix}, \qquad \boldsymbol{A}_{\mathrm{CB}} = \begin{pmatrix} \boldsymbol{I} & -[{}^{\mathrm{W}}\boldsymbol{d}]_{\times} \\ \boldsymbol{0} & \boldsymbol{I} \end{pmatrix}, \qquad {}^{\mathrm{W}}\boldsymbol{d} = {}^{\mathrm{W}}\boldsymbol{R}_{\mathrm{C}}\,{}^{\mathrm{C}}\boldsymbol{o}_{\mathrm{B}}, $$

and the base pose covariance is ${}^{\mathrm{W}}\boldsymbol{\Sigma}_{\mathrm{B}} = \boldsymbol{A}_{\mathrm{CB}}\,{}^{\mathrm{W}}\boldsymbol{\Sigma}_{\mathrm{C}}\boldsymbol{A}_{\mathrm{CB}}^{\top}$.

Proof. The base position follows from the camera pose as

$$ {}^{\mathrm{W}}\boldsymbol{p}_{\mathrm{B}} = {}^{\mathrm{W}}\boldsymbol{p}_{\mathrm{C}} + {}^{\mathrm{W}}\boldsymbol{R}_{\mathrm{C}}\,{}^{\mathrm{C}}\boldsymbol{o}_{\mathrm{B}} = {}^{\mathrm{W}}\boldsymbol{p}_{\mathrm{C}} + {}^{\mathrm{W}}\boldsymbol{d}. $$

Applying the perturbations gives

$$ \begin{aligned} {}^{\mathrm{W}}\tilde{\boldsymbol{p}}_{\mathrm{B}} &= {}^{\mathrm{W}}\tilde{\boldsymbol{p}}_{\mathrm{C}} + {}^{\mathrm{W}}\tilde{\boldsymbol{R}}_{\mathrm{C}}\,{}^{\mathrm{C}}\boldsymbol{o}_{\mathrm{B}} \\ {}^{\mathrm{W}}\boldsymbol{p}_{\mathrm{B}} + \delta\boldsymbol{p}_{\mathrm{B}} &= {}^{\mathrm{W}}\boldsymbol{p}_{\mathrm{C}} + \delta\boldsymbol{p}_{\mathrm{C}} + (\boldsymbol{I} + [\delta\boldsymbol{\theta}_{\mathrm{C}}]_{\times})\,{}^{\mathrm{W}}\boldsymbol{R}_{\mathrm{C}}\,{}^{\mathrm{C}}\boldsymbol{o}_{\mathrm{B}} \\ &= {}^{\mathrm{W}}\boldsymbol{p}_{\mathrm{C}} + \delta\boldsymbol{p}_{\mathrm{C}} + {}^{\mathrm{W}}\boldsymbol{d} + \delta\boldsymbol{\theta}_{\mathrm{C}} \times {}^{\mathrm{W}}\boldsymbol{d} \\ &= {}^{\mathrm{W}}\boldsymbol{p}_{\mathrm{B}} + \delta\boldsymbol{p}_{\mathrm{C}} + \delta\boldsymbol{\theta}_{\mathrm{C}} \times {}^{\mathrm{W}}\boldsymbol{d}, \end{aligned} $$

so, using $\boldsymbol{v} \times \boldsymbol{w} = -\boldsymbol{w} \times \boldsymbol{v}$,

$$ \delta\boldsymbol{p}_{\mathrm{B}} = \delta\boldsymbol{p}_{\mathrm{C}} - {}^{\mathrm{W}}\boldsymbol{d} \times \delta\boldsymbol{\theta}_{\mathrm{C}} = \delta\boldsymbol{p}_{\mathrm{C}} - [{}^{\mathrm{W}}\boldsymbol{d}]_{\times}\delta\boldsymbol{\theta}_{\mathrm{C}}. $$

The base orientation is ${}^{\mathrm{W}}\boldsymbol{R}_{\mathrm{B}} = {}^{\mathrm{W}}\boldsymbol{R}_{\mathrm{C}}\,{}^{\mathrm{C}}\boldsymbol{R}_{\mathrm{B}}$. The perturbed base orientation can be expressed using the convention of the lemma applied to the base by applying the camera perturbation with ${}^{\mathrm{C}}\boldsymbol{R}_{\mathrm{B}}$ fixed by the rigid mount as follows:

$$ \begin{aligned} {}^{\mathrm{W}}\tilde{\boldsymbol{R}}_{\mathrm{B}} &= {}^{\mathrm{W}}\tilde{\boldsymbol{R}}_{\mathrm{C}}\,{}^{\mathrm{C}}\boldsymbol{R}_{\mathrm{B}} \\ (\boldsymbol{I} + [\delta\boldsymbol{\theta}_{\mathrm{B}}]_{\times})\,{}^{\mathrm{W}}\boldsymbol{R}_{\mathrm{B}} &= (\boldsymbol{I} + [\delta\boldsymbol{\theta}_{\mathrm{C}}]_{\times})\,{}^{\mathrm{W}}\boldsymbol{R}_{\mathrm{C}}\,{}^{\mathrm{C}}\boldsymbol{R}_{\mathrm{B}} \\ &= (\boldsymbol{I} + [\delta\boldsymbol{\theta}_{\mathrm{C}}]_{\times})\,{}^{\mathrm{W}}\boldsymbol{R}_{\mathrm{B}}. \end{aligned} $$

Multiplying from the right by ${}^{\mathrm{W}}\boldsymbol{R}_{\mathrm{B}}^{\top}$ gives

$$ \boldsymbol{I} + [\delta\boldsymbol{\theta}_{\mathrm{B}}]_{\times} = \boldsymbol{I} + [\delta\boldsymbol{\theta}_{\mathrm{C}}]_{\times}, $$

so $\delta\boldsymbol{\theta}_{\mathrm{B}} = \delta\boldsymbol{\theta}_{\mathrm{C}}$. Stacking both results yields the linear map $\boldsymbol{A}_{\mathrm{CB}}$. □